DRONACHARYA
College of Engineering

DEPARTMENT OF CSE

DevMind AI: An AI-Powered Full-Stack Software Engineering Platform for Code Review, Repository Analysis, and Documentation

Submitted by:

Name: Barkha Yadav & Abhinav Thakur

College Roll No: 26039 & 26009

Branch: CSE

Duration: Aug – Nov 2026

Batch: Project-II (July–Dec 2026)

Submitted To:

Dr. Ashima Mehta, HOD CSE

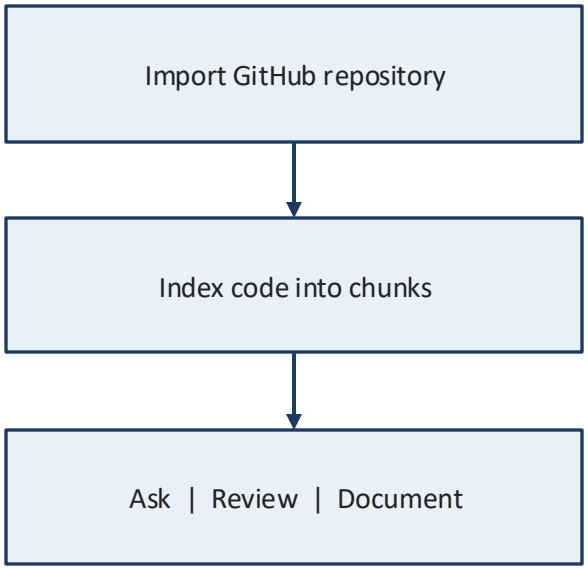
Project Guide:

Prof. Vimmi Malhotra

Introduction & Background

- DevMind AI is a web platform for understanding, reviewing and documenting source code, built as a React + Express application.
- Users sign in, connect their GitHub account, import a repository and let the system index its code.
- An LLM (Groq, with Gemini as fallback) then answers questions, reviews code and writes documentation using the indexed context.
- Reading existing code is a large part of software work; AI help is more useful when it is grounded in the actual repository.

Core idea



Problem Statement & Motivation

Understanding large codebases

New contributors spend a long time finding where features, routes and logic live.

Manual code review

Reviews are slow and depend on the reviewer; bugs and duplicate code can be missed.

Documentation gaps

READMEs and API docs are written by hand and often become outdated.

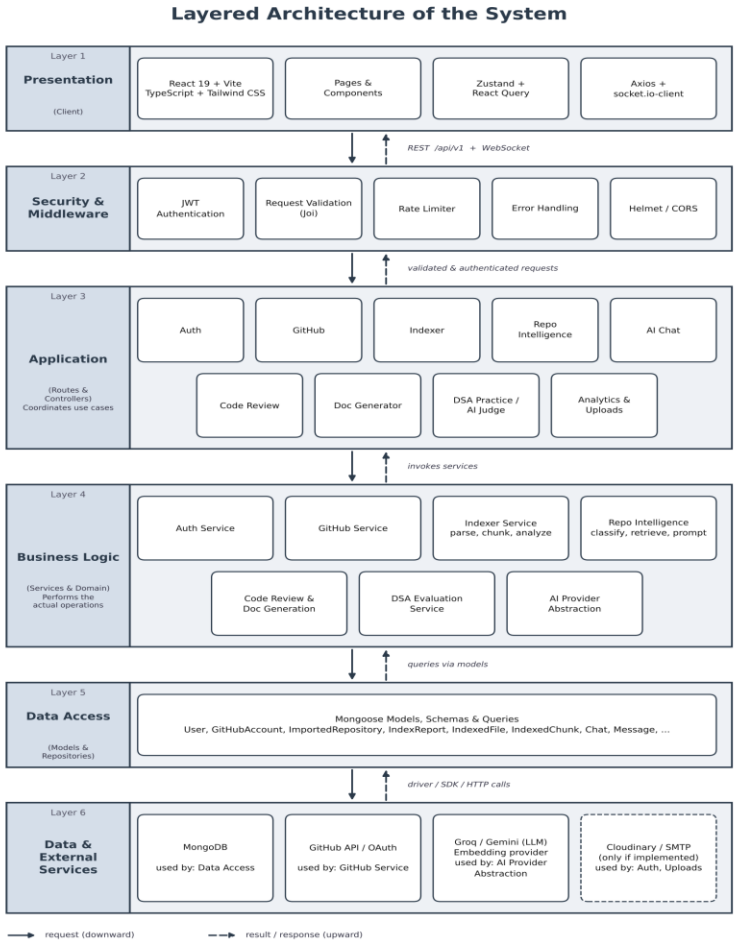
Scattered knowledge

Repository details are spread across files, commits and separate tools.

Motivation: a tool that works on the developer's own repository, instead of pasted snippets, can save time in understanding, reviewing and documenting code.

Objective Of DevMind AI

1. **AI-Powered Code Review:** To analyse source code using AI and identify potential bugs, security vulnerabilities, performance issues, and coding inefficiencies.
2. **Intelligent Repository Analysis:** To integrate with GitHub repositories and enable users to ask questions about their codebase and understand its structure and functionality
3. **Automated Documentation Generation:** To generate structured technical documentation, including README files, API documentation, installation guides, and architecture descriptions.
4. **Centralized Development Workspace:** To develop a unified full-stack platform that brings code analysis, repository understanding, and documentation generation together in a single, user-friendly environment.
5. **AI-Powered DSA Practice:** To provide an interactive coding practice environment where users can solve Data Structures and Algorithms (DSA) problems, execute code against test cases, receive AI-generated hints, analyse time and space complexity, and obtain feedback on their solutions.



LAYERED ARCHITECTURE

Layer 1 : is the Presentation layer, a React 19 + Vite + TypeScript client that handles pages, state and API calls and has no business rules.

Layer 2 : is Security & Middleware, which checks the JWT, validates input with Joi, applies rate limiting and handles errors, so bad requests are blocked early.

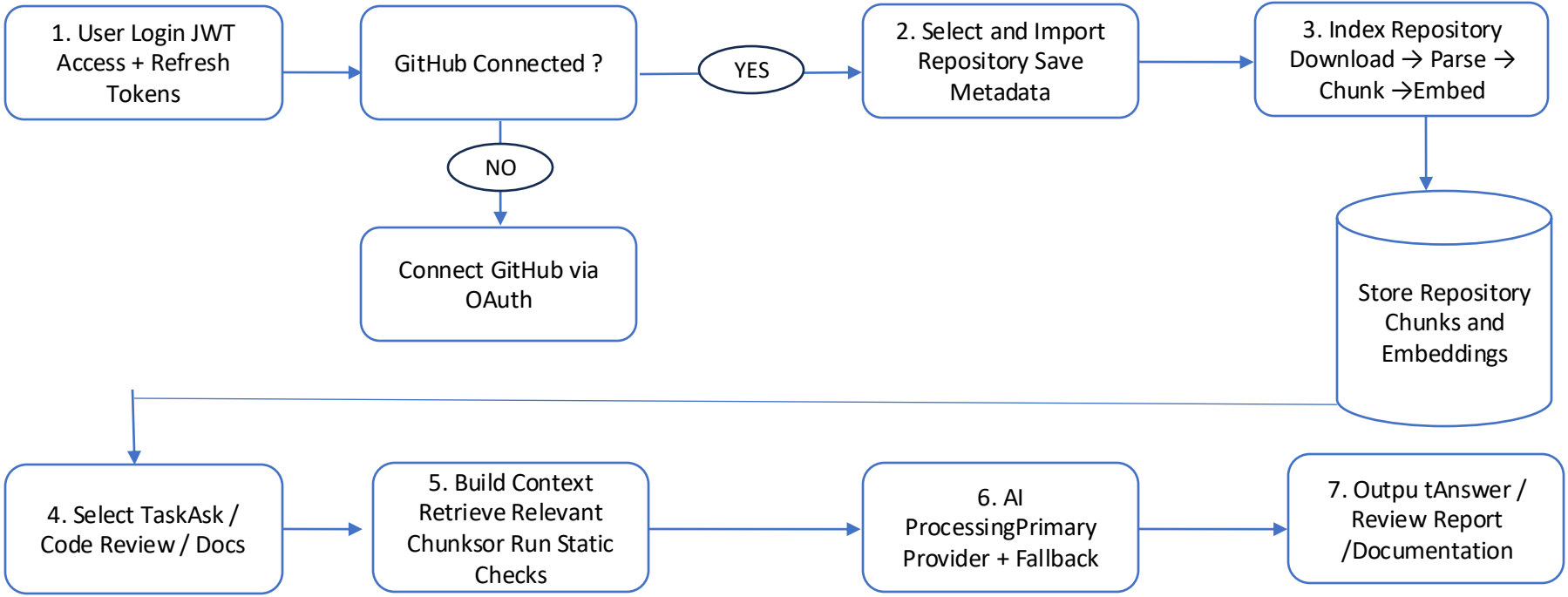
Layer 3 : is the Application layer, where controllers for each feature receive the request and pick the right service, coordinating the use case without processing it.

Layer 4 : is Business Logic, where services do the real work like indexing, code review, documentation and DSA evaluation, and the AI Provider Abstraction hides Groq and Gemini from the rest of the logic.

Layer 5 : is Data Access, made of Mongoose models and queries, and only this layer touches the database, so services never write raw queries.

Layer 6 : is Data & External Services, which includes MongoDB, the GitHub API and OAuth, GroqData Access, made of Mongoose models and queries, and only this layer touches the database, so services never write raw queries.

System WorkFlow

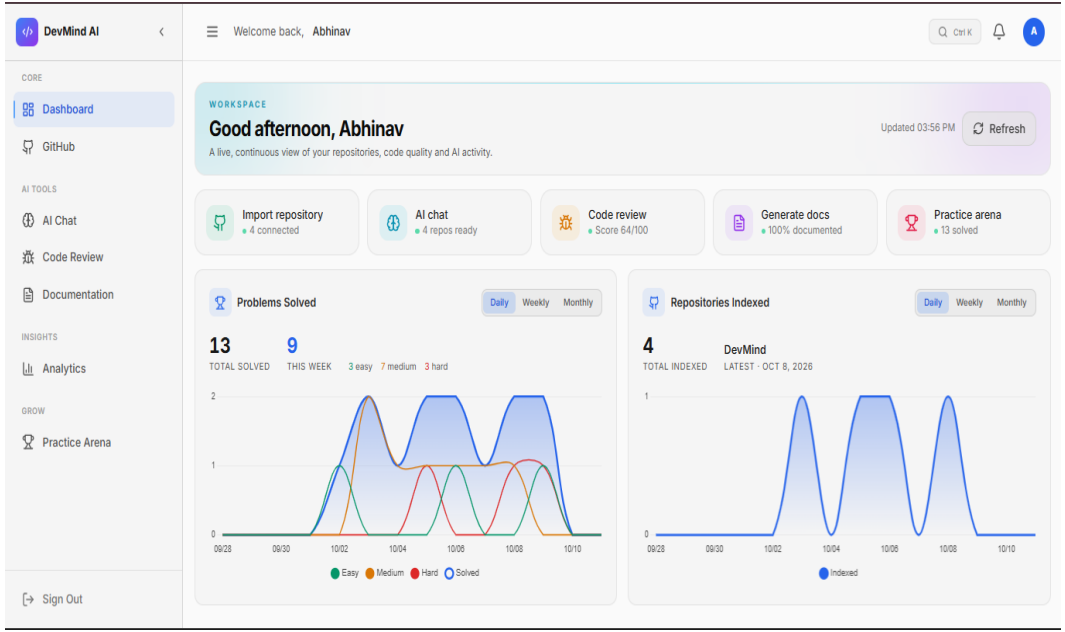


Major Modules & Key Features

Module	Implementation	Key feature
Authentication	JWT access + rotating refresh token, bcrypt, email verification	Secure login, password reset
GitHub Integration	OAuth with state check, Octokit REST client	Connect account, import repositories
Repository Indexer	Zipball download, parser, chunker, analyzer	Searchable code corpus, tech-stack summary
Repo Intelligence	Query classifier → retriever → prompt builder → LLM	Ask questions about code
AI Code Review	Complexity and duplicate checks + LLM review	Score, issues, refactoring suggestions
Doc Generator	Per-type prompts over indexed context	README, API docs, architecture, etc.
AI Chat	Chat sessions and messages in MongoDB; Socket.io notifications	Saved history, real-time notifications

Technology Stack & System Requirements

Frontend	React 19, TypeScript, Vite 6, Tailwind CSS, React Router
Backend	Node.js, Express 4, TypeScript
Database	MongoDB with Mongoose
AI	Groq (primary), Google Gemini (fallback)
GitHub	Octokit (REST API), GitHub OAuth
Real-time	Socket.io
Security	CORS, Joi validation
Deploy	Prettier, Render (render.yaml)



Outcomes, Future Scope & Conclusion

Outcomes

- Developed a full-stack AI-powered platform with JWT authentication and GitHub OAuth integration.
- Implemented GitHub repository import, code indexing, and chunk-based retrieval for codebase analysis.
- Enabled context-aware repository Q&A, AI-powered code reviews, and automated generation of 9 types of technical documentation.
- Integrated semantic and keyword-based code retrieval, AI provider fallback, and automated service-layer testing with Vitest.
- Configured Docker-based deployment infrastructure and Render hosting for application deployment.

Future Scope

- **Interview Arena:** Expand AI-powered DSA practice with timed coding sessions, automated code evaluation, progressive hints, AI-generated feedback, and performance analytics.
- **AWS S3 Integration:** Use Amazon S3 for scalable storage of repository archives, generated documentation, and exported reports, with secure access controls and lifecycle policies.
- **AI Evaluation and Reliability:** Introduce systematic evaluation of AI response accuracy, relevance, and consistency using test datasets and measurable quality metrics.
- **Repository-Based Recommendations:** Recommend relevant DSA problems based on programming patterns identified in users' GitHub repositories.

Conclusion: DevMind AI integrates AI-powered code review, repository understanding, and automated documentation into a unified full-stack platform. By retrieving relevant code context and providing it to an LLM, the system enables more contextual and efficient software development. Future improvements will focus on retrieval accuracy, AI response evaluation, and scalability.

.